



Autonomous assembly of structures using pinning control and formation algorithms

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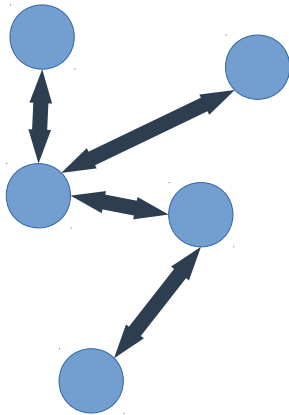
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Agenda

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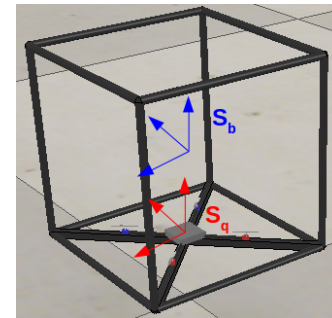
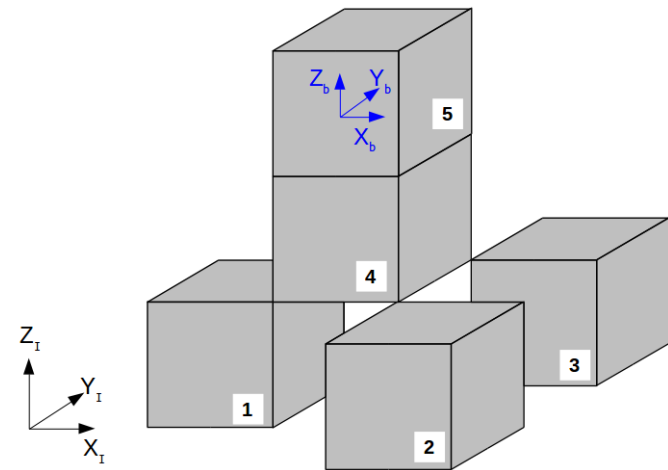
Introduction



- Systems with multiple agents are often interpreted as networks,
(agents = nodes and connections = edges)
- Network control methodologies are usually applied to
 - formation,
 - area coverage,
 - flocking or rendezvous
- We propose network-based algorithms to be applied at the autonomous construction of structures using self-actuated parts.

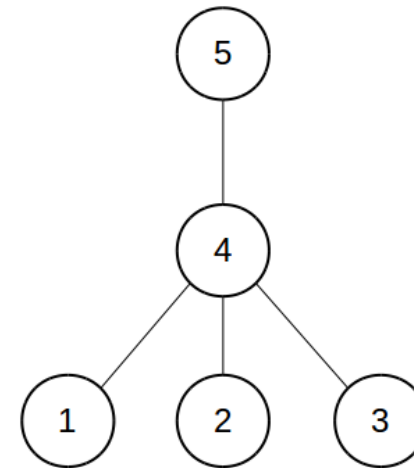
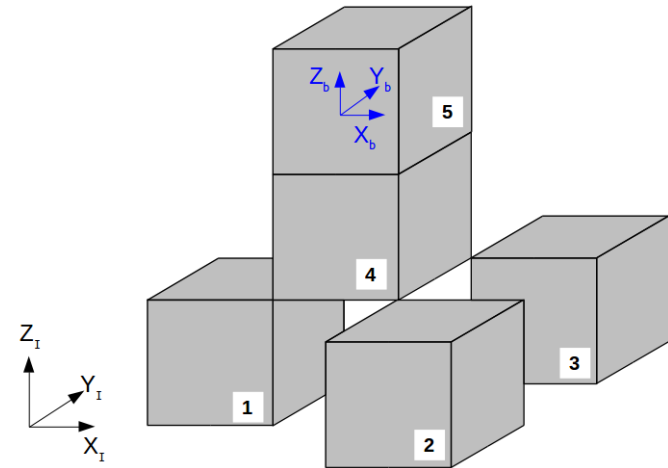
Problem description

- In autonomous construction systems, the structure could be:
 - Manipulated and transported by autonomous agents
 - The structure parts are self-actuated and move autonomously

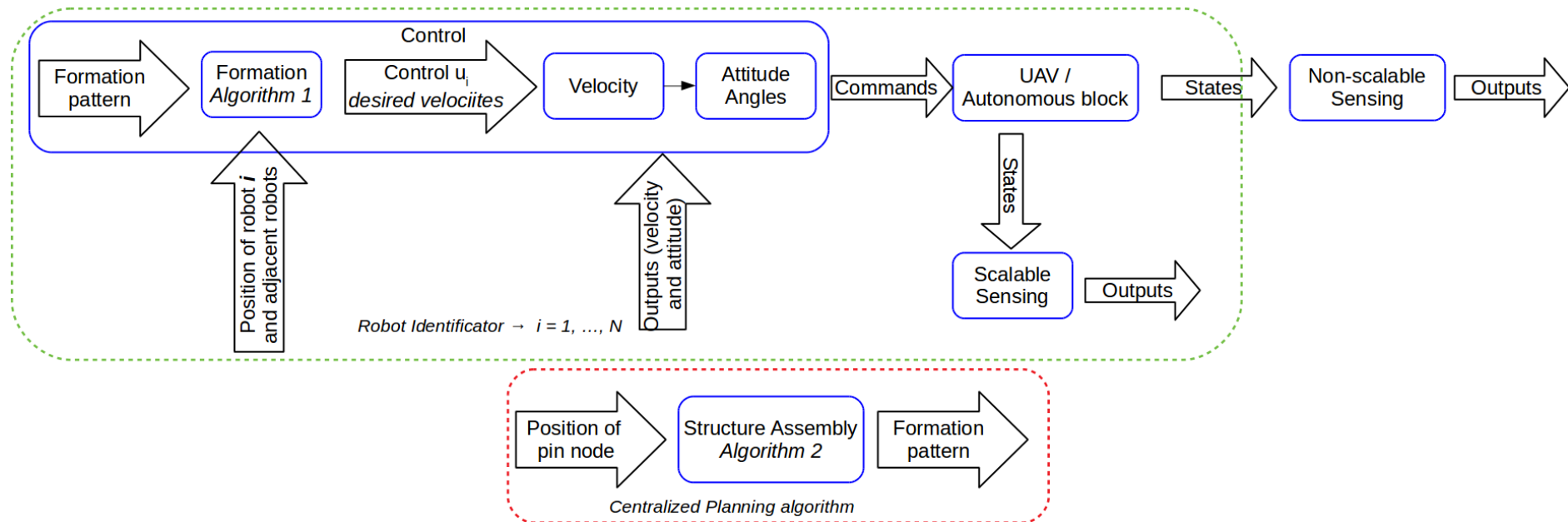


Problem description

- A network is represented by a graph $G = \{V, E, \mathbf{A}\}$
 - V – vertex set (nodes, agents)
 - E – edge set, connections between nodes
 - $\mathbf{A}$ – Adjacency matrix



Autonomous construction system - system architecture



Algorithms coordinate the movement of all agents:

- Algorithm 1 is a position control algorithm, being executed locally at each robot in order to achieve the desired location in a formation.
- Algorithm 2 is seen as a centralized planner, it has only one instance of execution during system operation, and it affects the desired position of all nodes

Autonomous construction system - system architecture

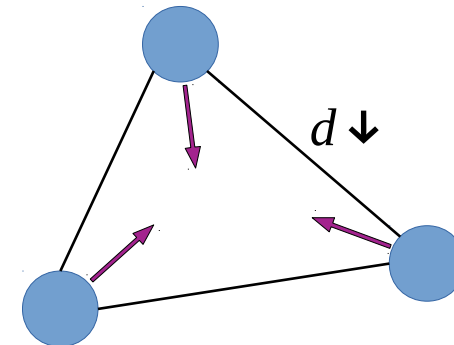
Algorithm 1 Formation

- 1: Get the actual position of the node (p_i)
 - 2: Get the actual position of the adjacent nodes (p_j)
 - 3: Get the desired position of the node (p_i^*)
 - 4: Get the desired position of the adjacent nodes (p_j^*)
 - 5: **if** i is a pin node **then**
 - 6: Get the desired final position of node i , p_r
 - 7: $h_i \leftarrow 1$
 - 8: **else**
 - 9: $h_i \leftarrow 0$
 - 10: **end if**
 - 11: Compute the desired velocity using equation (8)
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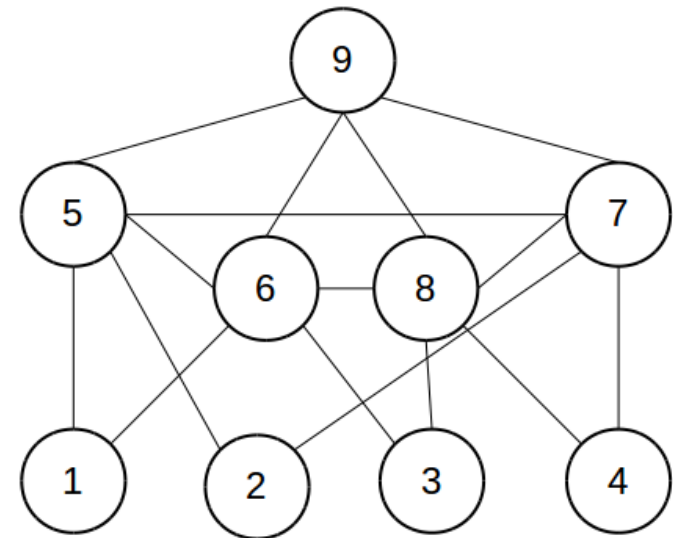
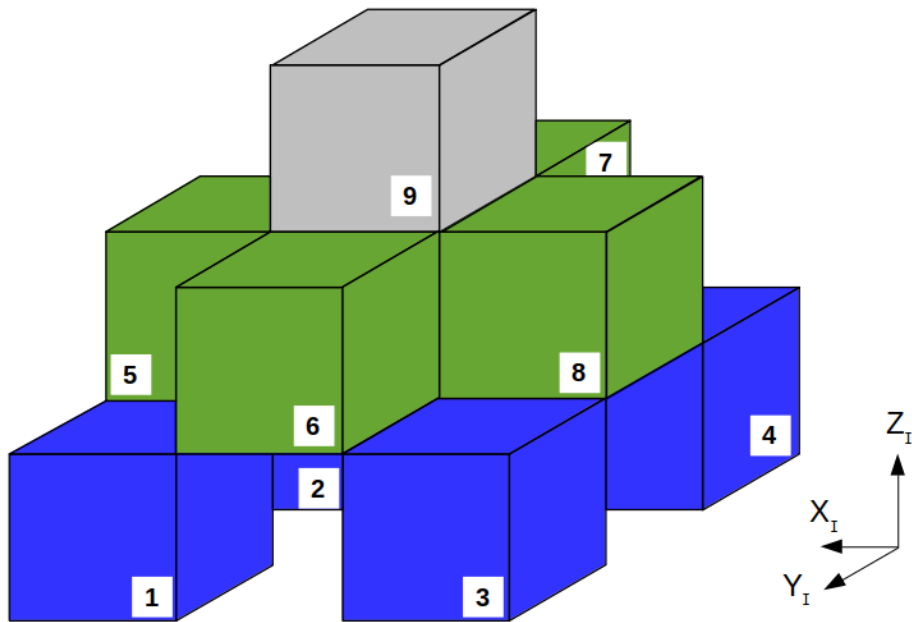
$$u_i = -h_i k_p (p_i - p_r) - k_f \sum_{j=1; j \neq i}^N a_{ij} [(p_i - p_j) - (p_i^* - p_j^*)]$$

Algorithm 2 Structure Assembly

- 1: Define the desired position of all nodes in the structure
 $p^* = \{p_1^*, p_2^*, \dots, p_N^*\};$
 - 2: Define the final position of node i , p_r
 - 3: $p^* = kp^*$ /* multiply the value of position in structure by a positive constant $k > 1$ to obtain the desired positions in the formation */
 - 4: **while** Structure not assembled **do**
 - 5: **if** Pinned node i reached desired position (p_r) **then**
 - 6: $p^* = kp^*$ /* multiply the desired position in the formation by a positive constant $k < 1$ to reduce inter-agent distance */
 - 7: Wait for nodes to get to the position
 - 8: **end if**
 - 9: **end while**
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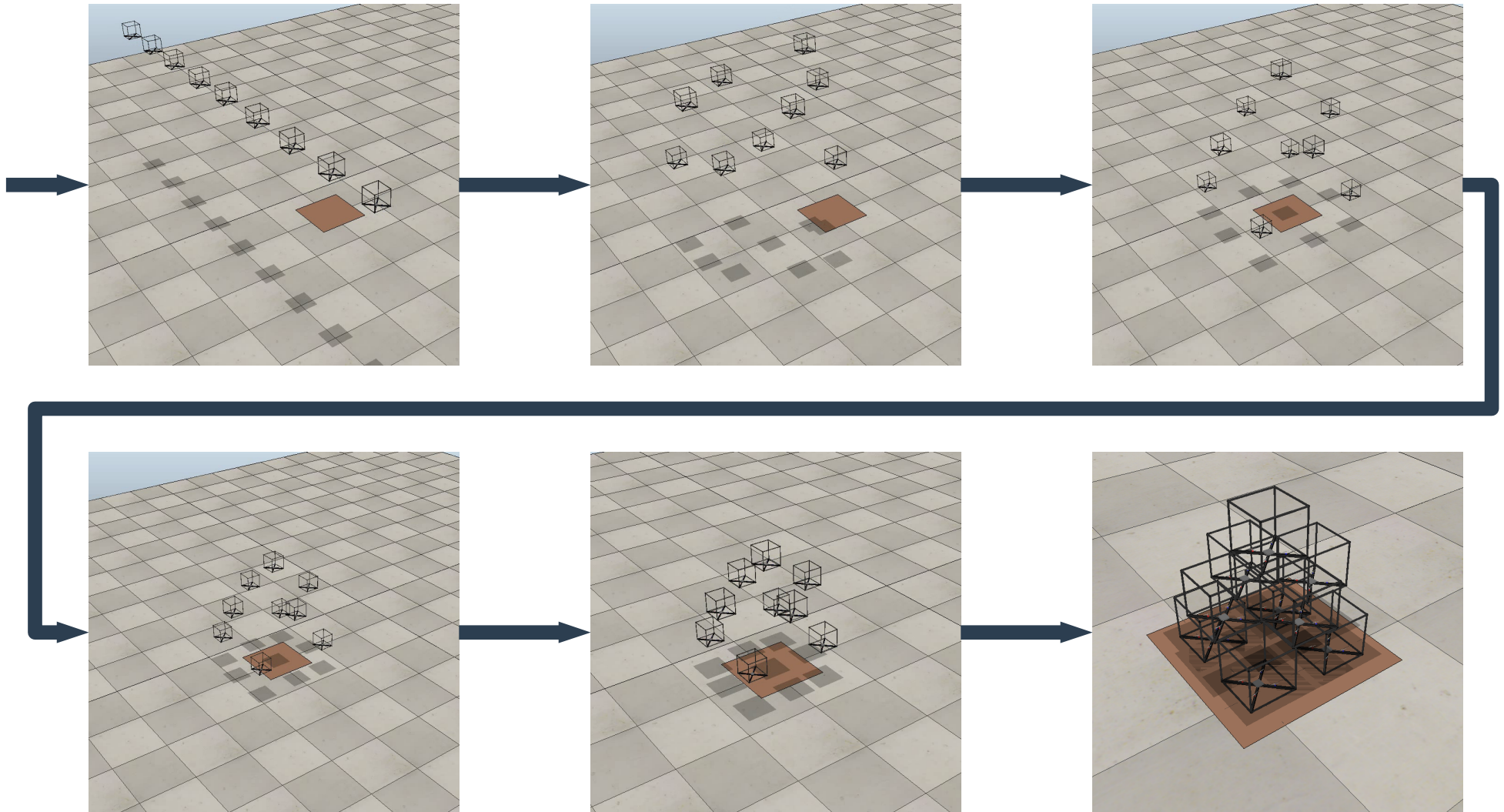


Simulations and results



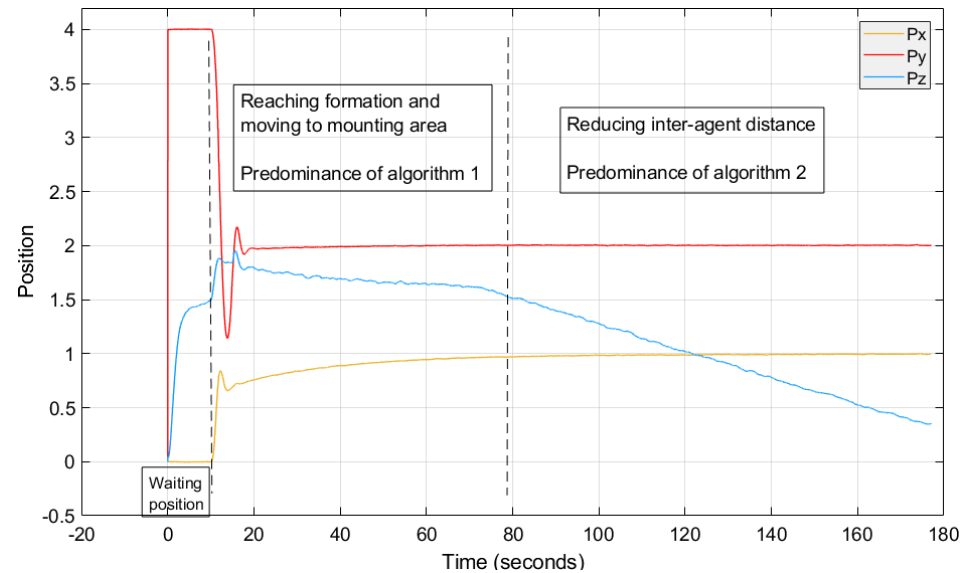
The topology is derived from the desired location of each block in the structure

Simulations and results



Selection of new pins

Investigating the position of the robot 9 (top of the pyramid), it is possible to see the actuation of each algorithm.



Conclusion

- We proposed control and planning algorithms that enable the autonomous construction.
- We modelled the multi-agent system as a network and the assembly of a structure as a formation problem.
- The inter-agent distance is gradually reduced in order to assemble the structure (contracting the formation size).
- A simulation trial demonstrated the effectiveness of our approach.
- In the future, we intend to explore priority on the mounting sequence using the proposed formulation.